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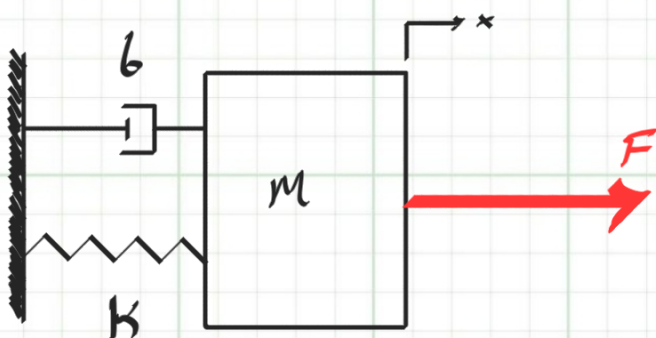
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Cases

*1	(M, b, K)	$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$
*2		$\begin{pmatrix} 1 & 1 & 2 \end{pmatrix}$
*3		$\begin{pmatrix} 1 & 2 & 1 \end{pmatrix}$
*4		$\begin{pmatrix} 2 & 1 & 1 \end{pmatrix}$
*5		$\begin{pmatrix} 0 & 1 & 1 \end{pmatrix}$
*6		$\begin{pmatrix} 1 & 0 & 1 \end{pmatrix}$
*7		$\begin{pmatrix} 1 & 1 & 0 \end{pmatrix}$

For the following cases Determine

* the order of the system (i.e. find the Laplace transform to infer order)

* if system is not a first or second order underdamped

System No hand written confirmation of values required

* Plot the step response (using Matlab)

* if 1st or 2nd order underdamped system is present then:

* Determine:

* Rise time (T_r)

* 2% settling time (T_s)

* Peak time (T_p)

* % overshoot

} Values to evaluate for
hand written calculation
Confirmation

Analysis

$$Ms^2 X(s) + Ds X(s) + KX(s) = F(s)$$

$$\Rightarrow \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Ds + K} = G(s)$$

Case #1

$$G_1(s) = \frac{1}{s^2 + s + 1}$$

* from 2nd general transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

⇒ Natural Frequency Damping Ratio

$$\omega_n = 1$$

$$\zeta = \frac{1}{2}$$

$$0 < \zeta < 1$$

⇒ Complex roots = $-\sigma \pm j\omega$
underdamped → $C(t) = Ae^{-\sigma t} \sin(\omega t)$

$$G_1(s) = \frac{1}{(s^2 + s + \frac{1}{4}) + \frac{3}{4}}$$

* from quadratic

$$= \frac{1}{(s + \frac{1}{2})^2 + \frac{3}{4}}$$

$$s = -\zeta\omega_n \pm j\omega\sqrt{\zeta^2 - 1}$$

⇒ Implied frequency shift,

$$\text{Note } * \mathcal{L}^{-1}(F(s+a)) = e^{-at} \mathcal{L}^{-1}(F(s)) = e^{-at} f(t)$$

$$\Rightarrow e^{-\frac{1}{2}t} \mathcal{L}^{-1}(F(s)) \Big|_{a=\frac{1}{2}} \\ F(s) = \frac{1}{s^2 + \frac{3}{4}}$$

$$\mathcal{L}^{-1}(F(s)) = \mathcal{L}^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{s^2 + (\frac{\sqrt{3}}{2})^2} \cdot \frac{2}{\sqrt{3}}\right) \because a F(s) = \mathcal{L}(f(t) a) \\ = \frac{2}{\sqrt{3}} \sin\left(\frac{\sqrt{3}}{2} t\right) \because \mathcal{L}^{-1}\left(\frac{\omega}{s^2 + \omega^2}\right) = \sin(\omega t)$$

$$\therefore f(t) = \frac{2 e^{-\frac{1}{2}t}}{\sqrt{3}} \sin\left(\frac{\sqrt{3}}{2} t\right)$$

* given that Case 1 is a 2nd order system and is underdamped T_r , $T_{s,2\%}$, T_p , + % overshoot are given as follows

$$T_r = \frac{1.64}{\omega_n}$$

$$T_{s,2\%} = \frac{4.0}{\xi \omega_n}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$$

$$\%OS = 100 e^{-\left(\frac{\pi \xi}{\sqrt{1-\xi^2}}\right)}$$

$$\xi = \frac{1}{2} \\ \omega_n = 1$$

$$\therefore \xi = \frac{1}{2} \mid \{0 < \xi < 1\} \Rightarrow \text{underdamped}$$

$\Rightarrow$

$$T_r = \frac{1.64}{1} = 1.64 \text{ sec}$$

$$T_{s,2\%} = 8 \text{ sec}$$

$$T_p = 3.6 \text{ sec}$$

$$\%OS = 16.3 \%$$

Case #2

given Case 2 is a 2nd order system and is underdamped as represented in Fig 2. Appendix A

$$G_2(s) = \frac{1}{s^2 + s + 2} = \frac{1}{(s + \frac{1}{2})^2 + (\frac{\sqrt{7}}{2})^2} \Rightarrow \mathcal{L}^{-1}(G_2(s)) = f_2(t) = \frac{2 e^{-\frac{1}{2}t}}{\sqrt{7}} \sin\left(\frac{\sqrt{7}}{2} t\right)$$

given the General Form 2nd order

$$\text{Given } 2\xi\omega = 1 \quad \omega = \frac{\sqrt{7}}{2}$$

$$\xi\omega = \frac{1}{2} \Rightarrow \xi = \frac{1}{\sqrt{7}} \because \xi = \frac{1}{\sqrt{7}} \mid \{0 < \xi < 1\} \rightarrow \text{underdamped}$$

$$\Rightarrow T_r \approx \frac{1.36}{\omega_n}$$

$$T_{s,2\%} = \frac{4.0}{\xi \omega_n}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$$

$$\%OS = 100 e^{-\left(\frac{\pi \xi}{\sqrt{1-\xi^2}}\right)}$$

$$\left\{ \begin{array}{l} \xi \approx 0.33 \\ \omega_n = \sqrt{7}/2 \end{array} \right.$$

$$\left\{ \begin{array}{l} \xi = 1/\sqrt{7} \\ \omega_n = \sqrt{7}/2 \end{array} \right.$$

$$T_r \approx 1.03 \text{ sec}$$

$$T_{s,2\%} = 8 \text{ sec}$$

$$T_p = 2.565 \text{ sec}$$

$$\%OS = 24.7 \%$$

* Compared to Values from Fig 2. Appendix A

Calculations confirmed with a reasonable margin of error.

Case #3

$$G_3(s) = \frac{1}{s^2 + 2s + 1} = \frac{1}{(s+1)^2} \Rightarrow \mathcal{L}^{-1}(G_3(s)) = e^{-t}t = f_3(t)$$

given the general 2nd Order system

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \Rightarrow \omega_n = 1 \text{ and } \zeta = 1$$

* Value of $\zeta = 1 \Rightarrow$ 2nd order system is critically damped
which is confirmed by plot structure in Fig 2 Appendix A

$\therefore$ Values for T_r , T_s , $\%OS$, T_P , $\%OS$ do not need to be confirmed

Case 4

$$G_4(s) = \frac{1}{2s^2 + s + 1} = \frac{1}{2} \left(\frac{1}{s^2 + \frac{s}{2} + \frac{1}{2}} \right) \quad * \frac{(s+\frac{1}{4})(s+\frac{1}{4})}{s+\frac{1}{2}+\frac{1}{16}}$$

$$\Rightarrow G_4(s) = \frac{1}{2} \frac{1}{(s+\frac{1}{4})^2 + \frac{7}{16}} \Rightarrow \mathcal{L}^{-1}(G_4(s)) = \frac{2e^{-\frac{1}{4}t}}{\sqrt{7}} \sin\left(\frac{\sqrt{7}}{4}t\right) = f_4(t)$$

$$2\zeta\omega_n = \frac{1}{2} \mid \omega_n = \frac{\sqrt{7}}{4} \Rightarrow \zeta = \frac{1}{\sqrt{7}}$$

* given Case 4 is a 2nd order underdamped system

$\Rightarrow T_r \approx \frac{1.36}{\omega_n}$	$\left\{ \begin{array}{l} \zeta \approx 0.33 \Rightarrow \omega_n = \sqrt{7}/4 \\ \zeta = 1/\sqrt{7} \\ \omega_n = \sqrt{7}/4 \end{array} \right\}$	$T_r \approx 2.036 \text{ sec}$	$\left. \begin{array}{l} * \text{ Compared to Values from Fig 2 Appendix A} \\ \text{Calculations confirmed with a reasonable margin of error.} \end{array} \right\}$
$T_{s, 2\%} = \frac{4.0}{\zeta\omega_n}$		$T_{s, 2\%} = 16 \text{ sec}$	
$T_P = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$		$T_P = 5.13 \text{ sec}$	
$\%OS = 100e^{(-\pi\zeta/\sqrt{1-\zeta^2})}$		$\%OS = 27.7 \text{ sec}$	

Case 5

$$G_5(s) = \frac{1}{s+1} \Rightarrow \mathcal{L}^{-1}(G_5(s)) = e^{-t} = f_5(t)$$

* Case 5 is a 1st order system * 1st Order systems can't be damped

$$\therefore T_r = \frac{2.2}{a} \mid a=1 \Rightarrow T_r = 2.2 \text{ sec.}$$

$$T_{s, 2\%} = \frac{4}{a} \mid a=1 \Rightarrow T_{s, 2\%} = 4 \text{ sec.}$$

$$T_P \rightarrow \infty$$

$$\%OS = 0$$

Case 6

$$G_6(s) = \frac{1}{s^2 + 1} \Rightarrow \mathcal{L}^{-1}(G_6(s)) = \sin(t) = f_6(t)$$

$$\because \omega_n = 1 \Rightarrow 2\zeta\omega_n = 0$$

$$\therefore \zeta = 0$$

* Case 6 is a 2nd order underdamped system

$\therefore$ Values for T_r , T_s , $\%OS$, T_p do not need to be confirmed

Case 7

$$G_7(s) = \frac{1}{s^2 + s} \Rightarrow \mathcal{L}^{-1}(G_7(s)) = 1 - e^{-t} = f_7(t)$$

* When finding poles of $G_7(s)$

$$s = -1$$

* Case 7 is a 2nd order and is unstable $\therefore G_7(s)$ has 1 real pole $\Rightarrow$ unstable system

$\therefore$ Values for T_r , T_s , $\%OS$, T_p do not need to be confirmed

Finishing Remarks

* Note: $y(t) = \mathcal{L}^{-1}\{G(s)X(s)\} | X(s) = \frac{1}{s}$

The full step response $y(t)$ is not required to solve for T_p , T_r , T_s , $\%OS$

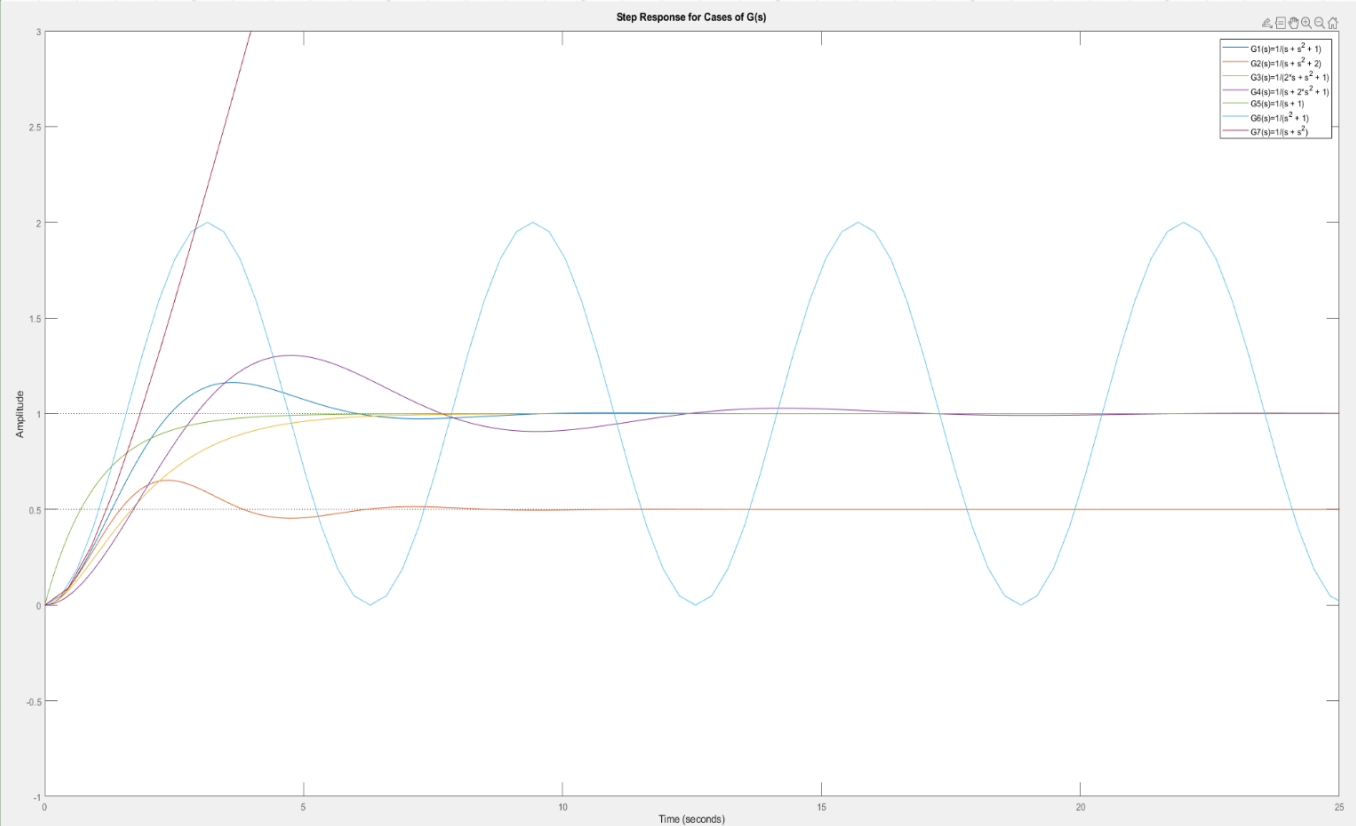
this is due to variable characteristics of the general forms of 1st & 2nd order systems

* Values for T_p , T_r , T_s , $\%OS$ found for cases that were either 1st or 2nd order underdamped systems are not exactly the same this is due to equations for T_r , T_p , T_s , $\%OS$ are given based on approximate use of ζ or have simplified numerical values in the numerator of equations used

$\therefore$ hand written calculated values validate the values gathered from the matlab models within a reasonable margin for error if values from both methods are rounded accordingly values approximately match.

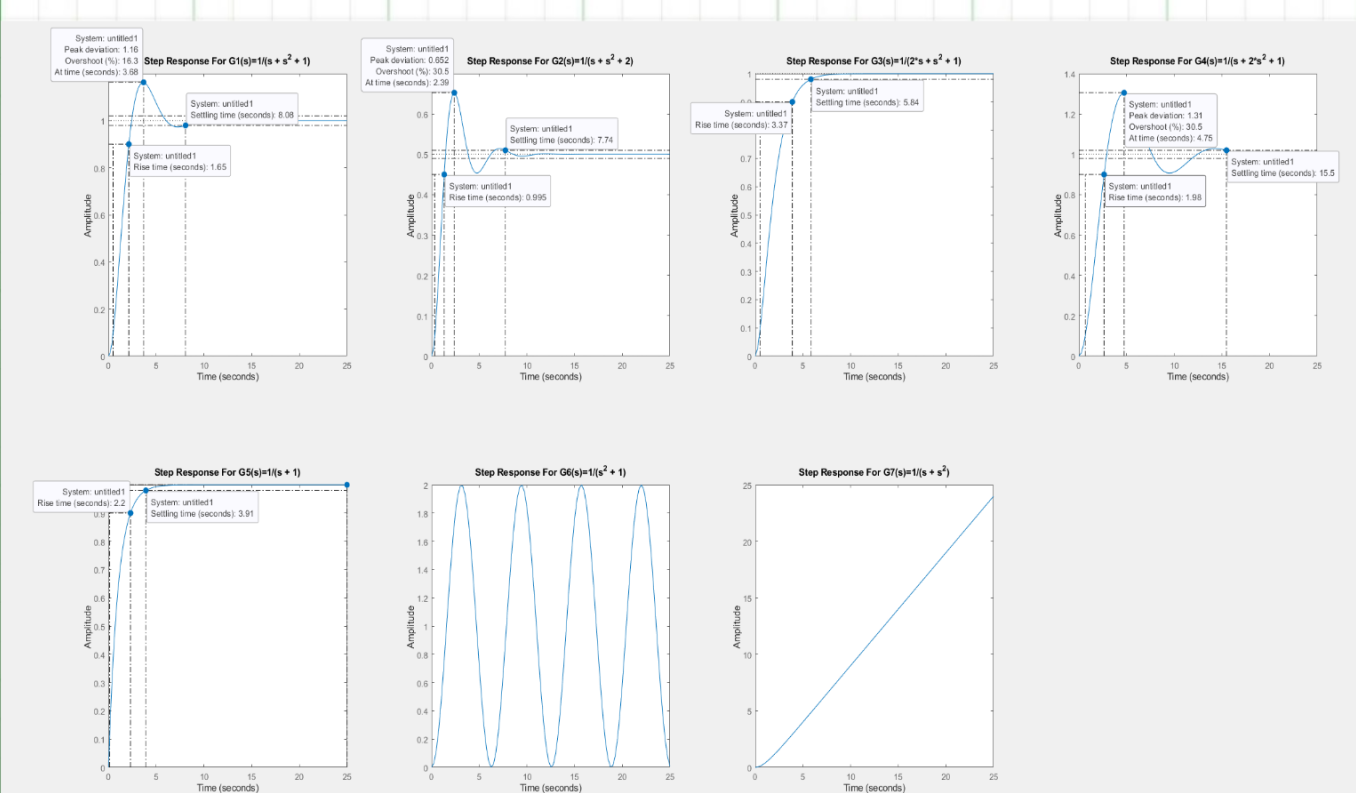
Appendix A

Fig.1



* Fig 1 shows all cases of $G(s)$ transformed to step response functions

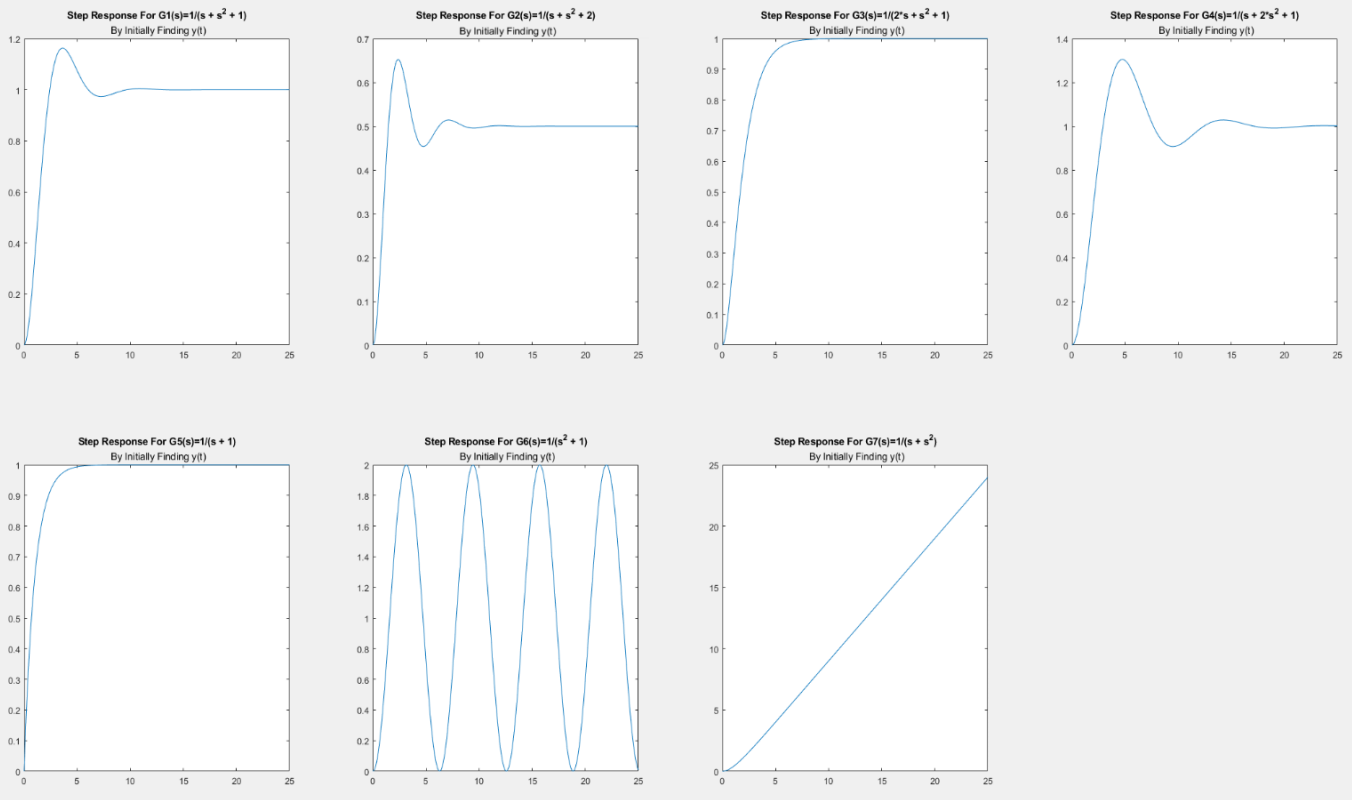
Fig.2



* plots generated using matlab 'step()' function

Appendix A

Fig 3



* plots in this figure are generated based on $y(t)$ equations generated for each case to validate the plots in fig 1 & 2 which are based on the matlab 'step()' function

Appendix B

```

1 clc
2 clear
3 clf
4 %%
5 %{
6 first define Values of M, b, k in a matrix for use in For loop to calculate
7 G(s) transform function for each case.
8 %}
9
10 M=[1 1 1 2 0 1 1]';
11 b=[1 1 2 1 1 0 1]';
12 k=[1 2 1 1 1 1 0]';
13 sys={};
14 syms s;
15
16 % Define unit step function to solve for y(t) for verification of plot generated by
17 % use of step() in figures 1 and 2.
18 X_s=1/s;
19 figure(1)
20 for j=1:7
21     % Define G(s) for each case
22     sys{j}=tf(1,[M(j) b(j) k(j)]);
23
24     % Converts G(s) for each case into character string for use in easy to
25     % read legend and titles
26     [Num,Den]=tfdata(sys{j});
27     Sym_sys_poly=poly2sym(cell2mat(Num),s)/poly2sym(cell2mat(Den),s);
28     Sym_2_cell=sym2cell(Sym_sys_poly);
29     Cell_2_str_sys{j}=string(Sym_2_cell);
30     Legend_str{j}='G'+string(j)+'(s)='+Cell_2_str_sys{j};
31
32     % Defines Y(t) For use in verification of step plots
33     y{j}=ilaplace(Sym_sys_poly*X_s);
34
35     % Sets Axis to fix limits to have optimal viewing wido of plot
36     axis([0 25 -1 3]);
37
38     % Plots step response for each case system
39     step(sys{j});
40     hold on
41
42     fprintf('\nGiven G%s(s)=%s the step response generates the following:\n',char(num2str(j)),Cell_2_str_sys{j})
43
44     % Determines Ts,2% Tr,Tp, & %OS for each system
45     stepinfo(sys{j})
46
47     %Determines roots, poles and, residual constants for each system
48     % used for derterming whether system is under-damped, over-damped,
49     % critically-damped, or un-damped
50     [r,p,c]=residue(1,[M(j) b(j) k(j)])
51
52 end
53 title('Step Response for Cases of G(s)')
54 legend(Legend_str{:})
55
56 %Create individual plots for each system
57 figure(2)
58 for j=1:7
59     subplot(2,4,j)
60     step(sys{j},25)
61     title('Step Response For '+Legend_str{j})
62 end
63
64 % Creates plots based on y(t)=G(S)X(s) for visual verification.
65 fprintf('The following equations are derived from: y(t)=G(S)*X(s)\n')
66 figure(3)
67 for j=1:7
68     t=0:0.1:25;
69     subplot(2,4,j)
70     plot(t,subs(y{j}))
71     title('Step Response For '+Legend_str{j}, ' By Initially Finding y(t)')
72     y_str{j}=string(sym2cell(y{j}));
73     fprintf('\ny%s(t) = %s\n',char(num2str(j)),y{j})
74 end
75
76

```

$$x^{-1}(G(s)X(s))$$

Given $G_1(s)=1/(s + s^2 + 1)$ the step response generates the following:

ans =

struct with fields:

```
RiseTime: 1.6390
TransientTime: 8.0759
SettlingTime: 8.0759
SettlingMin: 0.9315
SettlingMax: 1.1629
Overshoot: 16.2929
Undershoot: 0
Peak: 1.1629
PeakTime: 3.5920
```

r =

```
0.0000 - 0.5774i
0.0000 + 0.5774i
```

p =

```
-0.5000 + 0.8660i
-0.5000 - 0.8660i
```

c =

```
[]
```

Given $G_2(s)=1/(s + s^2 + 2)$ the step response generates the following:

ans =

struct with fields:

```
RiseTime: 0.9887
TransientTime: 7.7420
SettlingTime: 7.7420
SettlingMin: 0.4536
SettlingMax: 0.6524
Overshoot: 30.4890
Undershoot: 0
Peak: 0.6524
PeakTime: 2.3947
```



```
r =  
  
    0.0000 - 0.3780i  
    0.0000 + 0.3780i
```

```
p =  
  
   -0.5000 + 1.3229i  
   -0.5000 - 1.3229i
```

```
c =  
  
[]
```

Given $G_3(s)=1/(2*s + s^2 + 1)$ the step response generates the following:

```
ans =  
  
struct with fields:  
  
    RiseTime: 3.3579  
TransientTime: 5.8339  
SettlingTime: 5.8339  
SettlingMin: 0.9000  
SettlingMax: 0.9994  
Overshoot: 0  
Undershoot: 0  
    Peak: 0.9994  
    PeakTime: 9.7900
```

```
r =  
  
    0  
    1
```

```
p =  
  
   -1  
   -1
```

```
c =  
  
[]
```

Given $G4(s)=1/(s + 2*s^2 + 1)$ the step response generates the following:

ans =

struct with fields:

```
RiseTime: 1.9775
TransientTime: 15.4841
SettlingTime: 15.4841
SettlingMin: 0.9071
SettlingMax: 1.3049
Overshoot: 30.4890
Undershoot: 0
Peak: 1.3049
PeakTime: 4.7894
```

r =

```
0.0000 - 0.3780i
0.0000 + 0.3780i
```

p =

```
-0.2500 + 0.6614i
-0.2500 - 0.6614i
```

c =

```
[]
```

Given $G5(s)=1/(s + 1)$ the step response generates the following:

ans =

struct with fields:

```
RiseTime: 2.1970
TransientTime: 3.9121
SettlingTime: 3.9121
SettlingMin: 0.9045
SettlingMax: 0.9993
Overshoot: 0
Undershoot: 0
Peak: 0.9993
PeakTime: 7.3222
```

r =

1

p =

-1

c =

[]

Given $G(s)=1/(s^2 + 1)$ the step response generates the following:

Warning: Simulation did not reach steady state. Please specify YFINAL if this system is stable and eventually settles.

> In DynamicSystem/stepinfo (line 74)

In untitled (line 45)

ans =

struct with fields:

RiseTime: NaN
TransientTime: NaN
SettlingTime: NaN
SettlingMin: NaN
SettlingMax: NaN
Overshoot: NaN
Undershoot: NaN
Peak: Inf
PeakTime: Inf

r =

0.0000 - 0.5000i
0.0000 + 0.5000i

p =

0.0000 + 1.0000i
0.0000 - 1.0000i

c =


```
[]
```

Given $G(s) = 1/(s + s^2)$ the step response generates the following:

Warning: Simulation did not reach steady state. Please specify YFINAL if this system is stable and eventually settles. ↙

> In DynamicSystem/stepinfo (line 74)

In untitled (line 45)

```
ans =
```

```
struct with fields:
```

```

    RiseTime: NaN
  TransientTime: NaN
    SettlingTime: NaN
    SettlingMin: NaN
    SettlingMax: NaN
    Overshoot: NaN
    Undershoot: NaN
        Peak: Inf
    PeakTime: Inf

```

```
r =
```

```

-1
 1

```

```
p =
```

```

-1
 0

```

```
c =
```

```
[]
```

The following equations are derived from: $y(t) = G(s) * X(s)$

→ $\mathcal{L}^{-1}(G(s)X(s))$

$$y_1(t) = 1 - \exp(-t/2) * (\cos((3^{1/2}) * t) / 2 + (3^{1/2} * \sin((3^{1/2}) * t) / 2)) / 3$$

$$y_2(t) = 1/2 - (\exp(-t/2) * (\cos((7^{1/2}) * t) / 2 + (7^{1/2} * \sin((7^{1/2}) * t) / 2)) / 7) / 2$$

$$y_3(t) = 1 - t * \exp(-t) - \exp(-t)$$

$$y_4(t) = 1 - \exp(-t/4) * (\cos((7^{1/2}) * t) / 4 + (7^{1/2} * \sin((7^{1/2}) * t) / 4)) / 7$$

$$y_5(t) = 1 - \exp(-t)$$

```
y6(t) = 1 - cos(t)
```

```
y7(t) = t + exp(-t) - 1
```

```
>>
```